Final Report: The Impact of Climate Change and Water Availability of Northeastern Colorado Farmers

NR 518: Climate Impacts & Risk Assessment

Water Demands Group

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Table of Contents

Abstract

Agriculture in Northeastern Colorado is concentrated within the South Platte River Basin and is highly dependent on irrigation systems sustained by snowmelt-driven surface and groundwater supplies. As climate change alters precipitation patterns and reduces mountain snowpack, farmers and rural communities in this basin face increasing vulnerability to climate-induced water scarcity. This report proposes an adaptive strategy centered on improving irrigation efficiency, conserving soil moisture, and expanding financial incentives through state programs. Together, these strategies aim to reduce exposure, lessen sensitivity, and strengthen adaptive capacity, supporting long-term agricultural resilience, economic stability, and equity within the South Platte River Basin.

Image of a dry Platte River sourced from [Nebraska TV]

Section 1. Background

Water Demands in Colorado

Each year, Colorado uses 1.7 trillion gallons of water across its municipal, industrial, recreational, and agricultural sectors (CSU, n.d.). Most of this water originates as precipitation that accumulates as snowpack, which replenishes rivers and aquifers throughout the state as it melts (Mullane, 2023). However, the distribution of water supply does not align with population centers or agricultural demand. While a majority of the population and irrigated farmland are found on the Eastern side of the Continental Divide, nearly 80% of the water supply falls West of the Divide (CSU, n.d.). This spatial imbalance has created a persistent gap between water availability and Colorado's growing demand, requiring a complex network of pumping infrastructure, reservoirs, and regulatory guidelines to ensure the equitable distribution of water across the state. Agriculture places the greatest demand on this system, accounting for 89% of Colorado's annual water use to irrigate crops and pasture lands for livestock (CSU, n.d.). As Colorado's population grows and intensive agricultural production continues, the demand for water from the Colorado River Basin is projected to surpass the available supply within the next 30 to 50 years (DOI & USBR, 2012). This pressure on Colorado's water management system is expected to intensify as climate change impacts water supply in the arid environment.

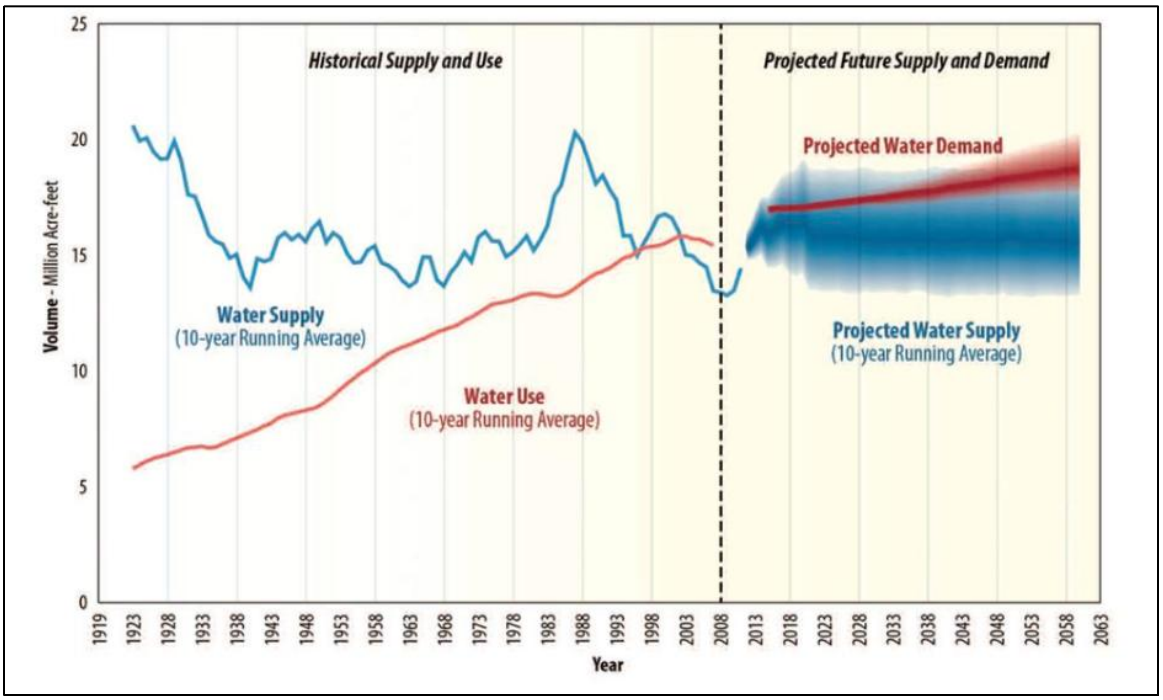

Figure 1. Historical and Projected Colorado River Water Supply and Demand.
This graph is sourced from research by the US Department of Interior and the Bureau of Reclamation (2012).

The Impacts of Climate Change on Water Availability

Climate change threatens water availability in Colorado through its effects on temperature, precipitation, and hydrologic processes. Climate models indicate that average temperatures are projected to rise in the state, from 2.5°F and 5°F by 2050 (Kennedy, 2014). Higher temperatures accelerate evaporation rates, intensify drought conditions, and increase water demands across municipal, industrial, and agricultural sectors (Hadjimichael, 2025). At the same time, climate change is altering precipitation patterns and reducing snowpack across Colorado, increasing uncertainty around both the timing and reliability of water supplies. Although future precipitation projections vary among climate models, there is strong agreement that a greater share of precipitation is falling as rain rather than snow, resulting in declining snowpack and earlier spring runoff (EPA, 2016; Bolinger et al., 2024). Snowpack currently supplies more than 70% of annual streamflow, yet snow water equivalent is projected to decline by 5-30% by mid-century (Bonnell, 2024; Bolinger et al., 2024). Figure 2 shows evidence of reduced snow water equivalent trends in Colorado with the black line referencing measurements taken in 2025. As depicted in the Figure, 2025 snow water equivalent levels in December are at 59% of the median (green) and tracking consistently with the minimum measurements from 1991-2020 (shown in red) (USDA & NRCS, 2025). As precipitation levels shift and the amount of snowpack continues to decline, the state is vulnerable to significant economic and social losses, especially within water-intensive sectors like agriculture.

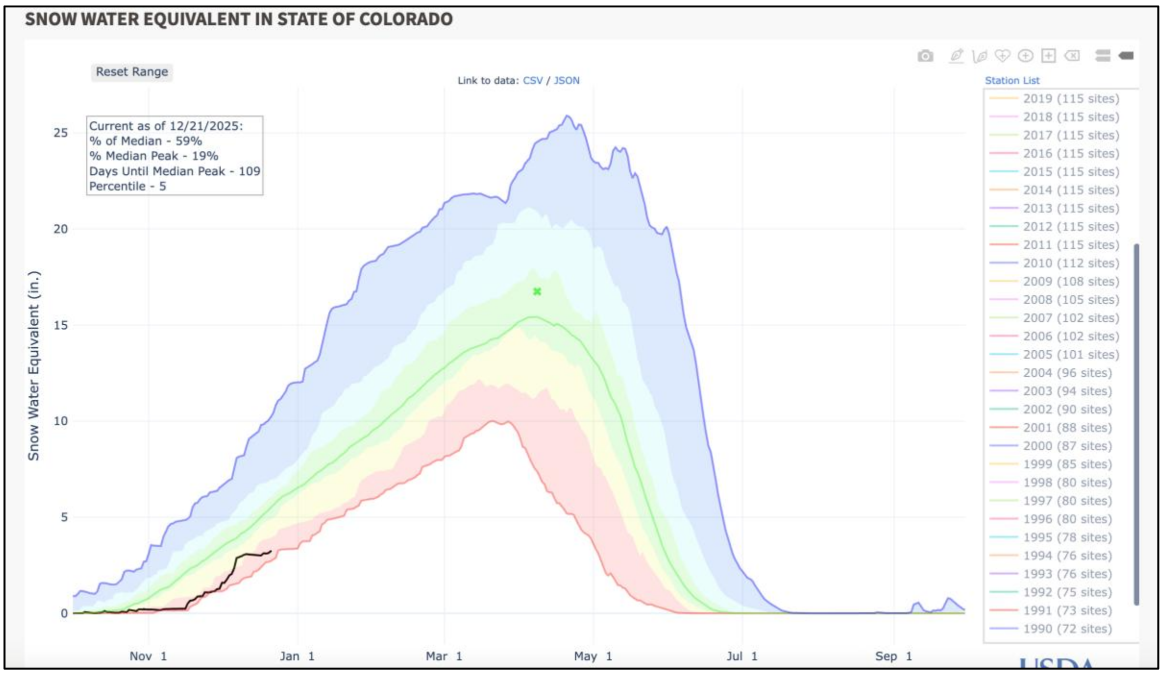

Figure 2. Snow Water Equivalent Measurements in Colorado, 1990-2025.
This graph is sourced from the USDA & NRCS (2025).

How Climate-Induced Water Scarcity Affects Colorado Agriculture

Agriculture is a foundational component of the physical, social, and economic landscape of Colorado, encompassing approximately 30 million acres, supporting more than 195,000 jobs, and generating $47 billion in economic activity for the state each year (Kelly, 2024). Because the agricultural system is the state's largest water consumer, water availability serves as a critical indicator of climate risk within this region (Water Education Colorado, n.d.). According to the North Central Climate Adaptation Science Center (2024), increasingly dry conditions are likely to reduce groundwater recharge, and when coupled with precipitation shifts and decreasing snowpack, these climate hazards pose a significant threat to irrigation dependency and water available for successful crop production. Shown in Figure 3, irrigated acres have decreased across the state from 1997 to 2022, especially in the South Platte Region of Northeastern Colorado. Along with reduced water availability, additional negative climate impacts on farmland necessitate the development of adaptive management strategies to promote long-term water sustainability and resilience.

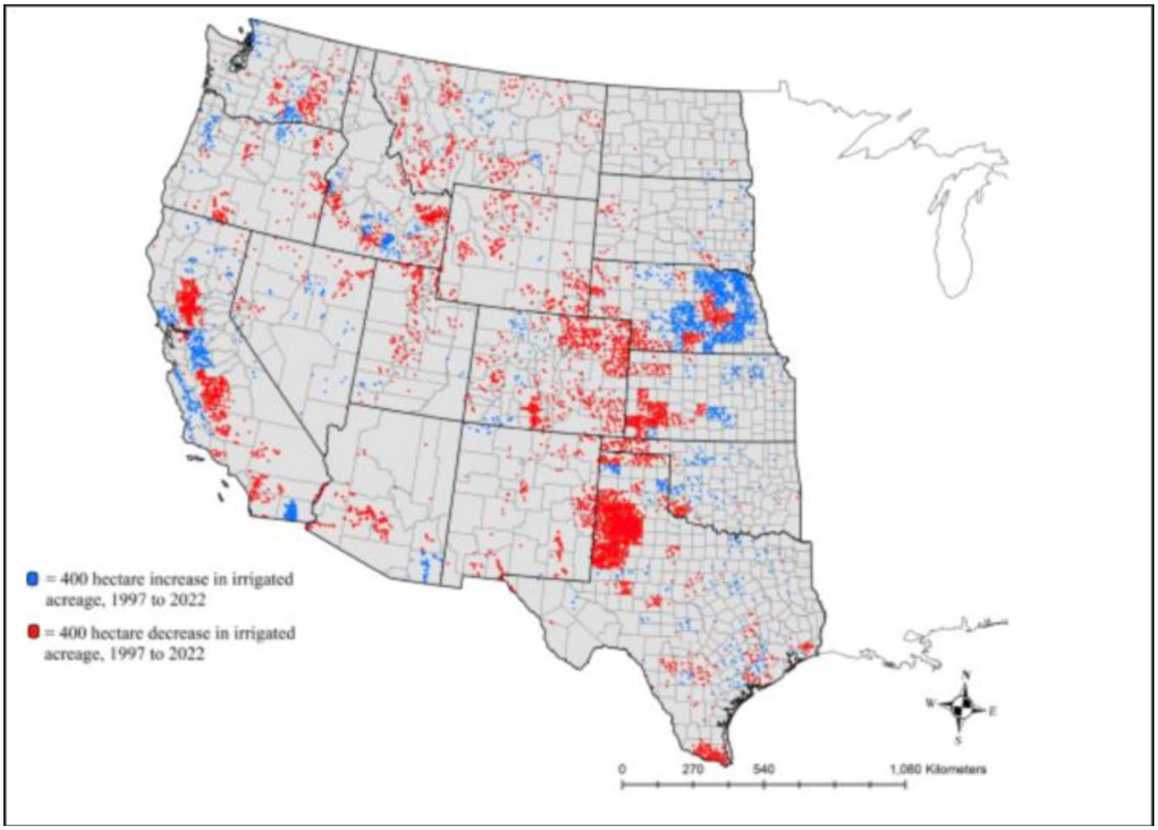

Figure 3. Shifts in Irrigated Land in the Western United States from 1977 to 2022. This graph is sourced from Hrozencik et al. (2025).

Climate-induced droughts, increasing temperatures, and insufficient irrigation contribute to drier soils, heightened prevalence of pest outbreaks, and production losses (NOAA & NIDIS, n.d.). Dry soils impact crop health by limiting the uptake of nutrients by crops and reducing organic matter decomposition (Al-Kaisi, 2017). Declining crop health can lead to high rates of production losses, straining economic and social systems in farming communities. To cover diminishing returns due to droughts, the USDA paid $5.6 billion to seven states within the Colorado River Basin between 2017 and 2023 (Schechinger, 2024). Though crop insurance is a vital support system, these production losses leave farmers, especially underserved minority farmers, financially vulnerable to future climate impacts. Furthermore, research from the NCCASC states that rural areas and farms using acequias, or snow-fed irrigation systems, will be disproportionately impacted by climate-induced water shortages, potentially leading to higher rates of food insecurity, more significant crop losses, and reduced well-being. Given agriculture's central role in Colorado's economy and food system, these risks highlight the need for urgent action to assess the impact of climate change on farmers and their land. Key opportunities for adaptive management include addressing soil health concerns and strategic reduction of on-farm water use to improve crop health and help farmers build resilience.

Section 2. Vulnerability Analysis

The vulnerability of agricultural systems in Northeastern Colorado is shaped by the interaction of environmental, social, and institutional factors that determine farmers' exposure, sensitivity, and adaptive capacity to climate-induced water scarcity. This region, located within the South Platte River Basin, depends heavily on snowmelt-fed irrigation systems and groundwater pumping to sustain crop production. As climate change alters the hydrologic cycle, these coupled water and agricultural systems face increasing stress that crop yields, economic stability, and the long-term viability of farming communities.

Exposure

Farmers in Northeastern Colorado experience high exposure to climate hazards driven by reductions in snowpack, earlier spring runoff, and declining groundwater recharge. Warming temperatures accelerate evapotranspiration, increasing water demand at the same time that water supplies are becoming more irregular. Earlier runoff reduces alignment between peak water availability and peak crop water needs, intensifying late-season shortages (Fassnacht & Pfohl, 2025). Groundwater pumping, often used as a buffer during drought years, is becoming less reliable as aquifers decline across the South Platte region (NFS, 2008).

Compounding these climatic pressures, competition for water from rapidly growing Front Range municipalities places additional strain on already limited surface water supplies. Water that historically supported agricultural use is increasingly diverted toward urban and industrial sectors, heightening the exposure of farmers who rely on consistent water deliveries to irrigate corn, alfalfa, and other high-demand crops.

Sensitivity

Sensitivity within the agricultural sector is high, primarily because crop production in Northeastern Colorado is dominated by water-intensive cropping systems that depend on reliable irrigation. Corn and alfalfa, which cover much of the irrigated acreage within the basin, require substantial amounts of water during the growing season and are particularly vulnerable to heat stress. Research shows that temperatures exceeding 30°C (86°F) during critical growth periods, such as tasseling and silking in corn, can significantly reduce yield, even when irrigation is available (Delgado et al., 2024). As warming accelerates, these stress events are projected to become more frequent and intense.

Farmers' sensitivity is further amplified by soil moisture decline, heightened drought stress, and increased pest pressures, all of which have been exacerbated by a warming climate. As soil dries out rapidly, their ability to retain water is diminished, reducing both crop resilience and irrigation efficiency (Andales & Schneekloth, 2017). Social and institutional factors also heighten sensitivity. Colorado's senior-junior water rights system disproportionately impacts newer, minority, and smaller-scale farmers, who are the first to face curtailments during drought (CSU, 2023). These inequities make many producers more vulnerable to water shortages, even in years when senior rights holders receive their full allocations.

Adaptive Capacity

Adaptive capacity in Northeastern Colorado is moderate but constrained by economic, institutional, and knowledge-based barriers. Farmers in this region have a long history of managing water scarcity through innovative practices, and many have adopted techniques such as no-till farming or diversified crop rotations to reduce water use and maintain soil health. Precision irrigation, particularly drip systems, has shown the potential to reduce water consumption by up to 60% and improve yields by as much as 90% (Chu, 2017; EPA 2025). However, initial installation costs, averaging around $3,000 per acre, pose a substantial financial barrier, particularly for small and mid-scale operations already experiencing drought-related economic losses (Chu, 2017).

Institutionally, adaptive capacity is limited by the absence of funding in several Northeast Colorado counties, leaving communities without the resources and support to create and execute climate-informed hazard mitigation plans (Solomon, 2024). Surveys conducted in this region reveal that 60% of farmers do not believe their community is prepared for climate change, indicating gaps in climate literacy, uncertainty about future water availability, and limited access to technical support. Without strengthened extension services targeted incentives, and equitable water management frameworks, adaptive capacity will remain insufficient to offset rising climate pressures.

Taken together, Northeastern Colorado exhibits high vulnerability to climate-induced water scarcity. Environmental exposure is intensifying due to declining snowpack and shifting hydrology, while agricultural sensitivity remains elevated because of reliance on water rights. Adaptive capacity, although improving in some areas, remains constrained by high economic costs, limited institutional support, and barriers to adopting innovative technologies. This analysis demonstrates the need for integrated, farmer-centered adaptation strategies that reduce total water demand, enhance soil water retention, and expand financial and technical support systems. Strengthening adaptive capacity will be essential to maintaining agricultural productivity, rural livelihoods, and long-term resilience across the South Platte River Basin.

Section 3. Adaptive Strategy Design

Objectives

The adaptative strategies presented in this section are directly informed by the vulnerability analysis, which identified high exposure to declining snowpack and altered runoff timing, high sensitivity of water-depending cropping systems, and moderate but constrained adaptive capacity among farmers in this South Platte River Basin. In response, the adaptive strategy outlined in Figure 4 prioritizes interventions that reduce exposure by lowering total irrigation demand, decrease sensitivity by improving soil water retention and crop resilience, and strengthen adaptive capacity through monitoring, learning, and stakeholder engagement. Precision irrigation and soil health improvements were selected because they directly address the primary drivers of vulnerability identified in the previous section while remaining feasible within existing agriculture systems and institutional frameworks.

Over the course of 5 years, from 2025 to 2030, these strategies will be monitored for their effectiveness in reducing on-farm water use and improving soil health. Working closely with stakeholders throughout the monitoring process, the aim of this adaptive management program is to learn from seasonal data and adapt strategic plans to ensure they best support the values and needs of farmers facing climate change uncertainties. Key stakeholders include irrigated crop producers, who rely on consistent and affordable water supplies to sustain operations, and irrigation districts which manage water delivery and influence on-farm irrigation practices. State and local agencies, such as the Colorado Water Conservation Board, drought planning, conservation incentives, and data collection efforts are also relevant. Colorado State University Extension and local conservation districts play a critical role in technical assistance, farmer education, and practice implementation.

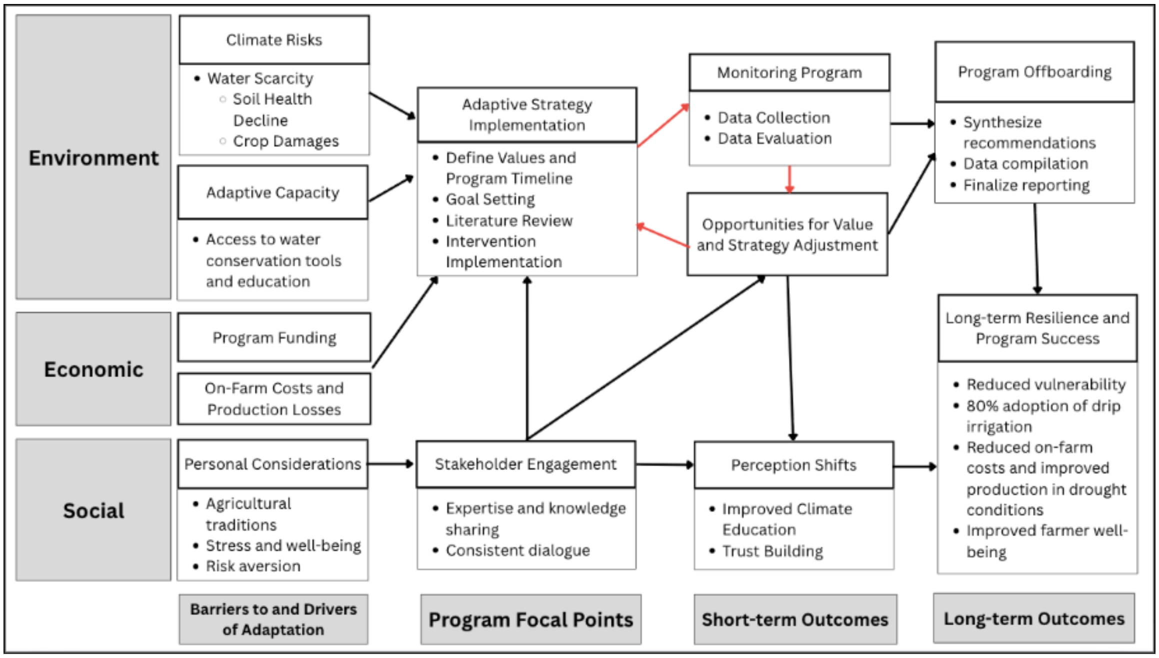

Figure 4. Conceptual Figure of Adaptation Management and Monitoring for Climate-Induced Water Scarcity on Farms in the South Platte Region.

Since water is a shared public resource, we created a strategy that prioritizes people, profit, and the planet. As seasonal snowpack and downstream flows are altered, water becomes a highly stressed commodity with varying needs. This scarcity decreases soil health and subsequently damages crops as metabolic processes are halted. Our adaptive management plan includes the introduction of drip irrigation and soil amendments to preserve water supplies and enhance soil composition to promote growth.

Similarly, fertilizer advancements, seed modifications, and land management practices may attempt to keep up with crop sensitivities but are not comprehensive as delayed growth and harvest patterns persist. These remediations can be costly as farmers bear the economic burden, especially when paired with production losses. We forecasted a future where smarter water and soil management would minimize the need for additional costs as a more secure water source becomes available and the amendments correct nutrient concentrations through biogeochemical processes. Additionally, our project goal explains the relationship between rural communities in Northeastern CO and agriculture. With the current state of the industry, livelihoods are threatened by additional strain to make ends meet and maintain the ag legacy. Our plan addresses these concerns with the proposed technologies and educational opportunities, easing financial and cultural stress as production rates stabilize.

Proposed Interventions

Soil amendments enhance desirable soil characteristics like nutrient content, texture, and fertility. Compost, specifically, is a mix of decomposed organic materials like food waste and yard scraps that have been heated and fully incorporated together. It is rich in nutrients, encouraging microbial processes that enhance water holding capacities and reduce erosion. Similarly, biochar is created when organic waste materials are set on fire in low-oxygen conditions. The resulting product increases soil pH to make nutrients more soluble, filters polluted soil, lowers GHG emissions, and sequesters atmospheric carbon. Both amendments would increase adaptive capacity for farmers as they improve soil health for higher quantity and quality crops.

This is especially true when paired with proper water allocation on the field. Drip irrigation delivers water directly to the root of the plant, increasing the resource's efficiency on the field. This approach reduces the total amount of water used with less evaporation and runoff potential, improves crop health as moisture levels remain consistent, and enables targeted distribution of nutrient additives. Together, these two solutions will ensure our short-term goals for 2030 are met with reassurance given through the return in yields (estimated up to 90%) and water saved (around 60%) (Chu, 2017; EPA 2025). Figure 5 illustrates our proposed monitoring plan to evaluate the effectiveness of these interventions with flexibility mechanisms to ensure seasonal adjustments are possible based on the resulting quantitative and qualitative data.

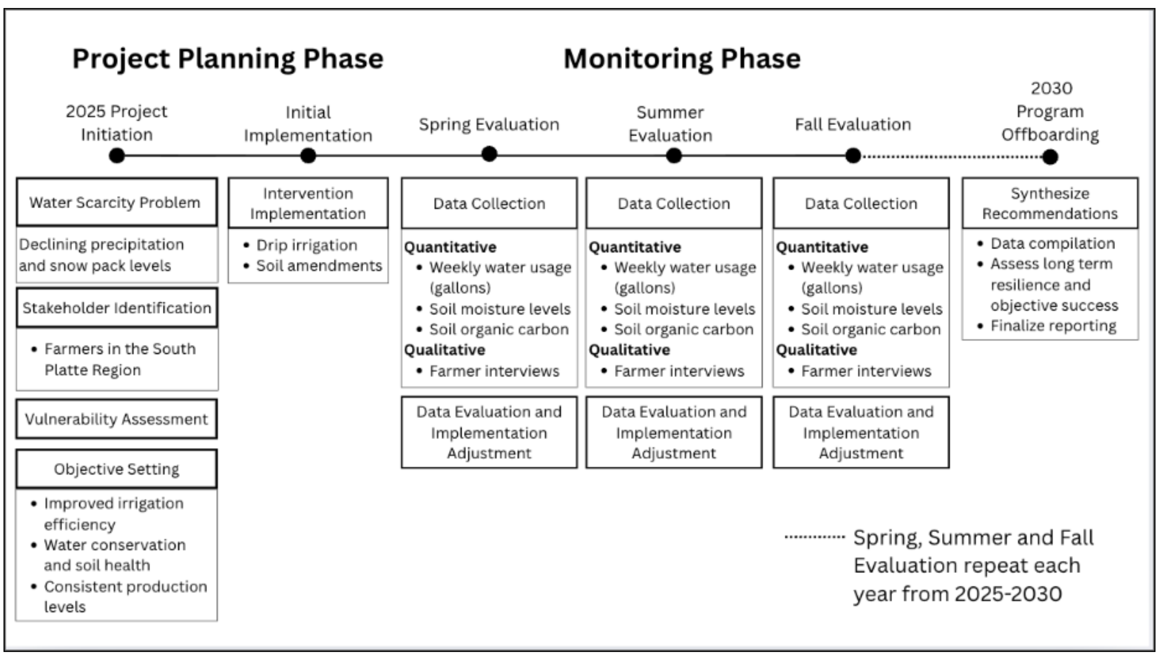

Figure 5. Monitoring Process for Reducing Agricultural Vulnerability to Water Scarcity

Improving soil health is key to reducing vulnerability and adapting to climate change for farmers in Northeastern Colorado. Good water retention in the soil allows these stakeholders to be more resilient towards drought and lower water supply. This adaptive management strategy is designed to reduce agricultural vulnerability stemming from water scarcity by linking monitoring directly to management decisions. The mechanism for adaptation is grounded in improving soil water retention and irrigation efficiency, which reduces crop sensitivity to declining water supplies. While implementation challenges exist, including high upfront costs for drip irrigation systems, access to soil amendments, and technical knowledge requirements, these strategies have high potential to improve outcomes by lowering water use and improving soil resilience.

Section 4. Summary

Through our research, it is clear that this project is incredibly important and necessary for the future of Colorado livelihoods. Especially in Colorado's arid and dry environment, where water is already scarce, the future of climate change poses serious threats to our water supply. In order to ensure the future stability of Colorado food systems and food security, it is essential that the resilience and health of Colorado farms are ensured now. Currently, Colorado farmers are incredibly vulnerable to decreased water supply because it directly impacts their ability to grow crops. By increasing the health of field soils and implementing more efficient watering techniques, vulnerability can be reduced. The primary reason why our adaptive strategy management plan and monitoring plan works is because it utilizes site-specific management decisions in order to ensure that each farm's needs and goals are met. There is not one solution to this problem; it will take various combinations of different management practices for each specific farm to achieve greater resilience. Furthermore, our strategies work because they prioritize the health of the soil in order to improve the water holding capacity and resilience of the soil. This translates to improved resiliency and stability of the overall farm.

This project should be funded for a variety of reasons. First, the stability of farms and food systems is vital to the survival of humans. Farmers have extremely narrow profit margins and cannot afford to make different management decisions. This is partially due to the fact that there is a lot of risk involved as yields may be temporarily affected during the adjustment process to new management decisions. Also, smaller and mid-scale farms do not have the same level of income that large-scale farmers do, making funding even more important. This is why outside investment is vital to the success of our management plan as farmers cannot be expected to make and fund these changes on their own. Over time, farmers will experience decreased vulnerability and have more stable operations in the future to repay investors and teach farmers in similar regions how to adapt to their water stressed conditions.

As far as the main outcomes this project will achieve, there are many. Most importantly, it will allow Colorado farmers to maintain their livelihood here and not be forced to move elsewhere. This is important because there are a significant portion of Colorado counties that are rural and agricultural communities. Without this livelihood, many communities would be lost. Especially in our region of interest, the South Platte River Basin, where the communities are primarily agricultural. Another main outcome of this research would be to ensure that our agricultural system is protected the best it can be against climate change. In the future, there will be other environmental threats other than just decreased water supply, so it is vital that the health and resiliency of farms is protected now. As mentioned earlier, a stable food system is foundational to the success of humans. So, mitigation and adaptation measures taken now will be hugely valuable in the future. The adoption of this project will have long-term regional benefits that can be scaled throughout the country.

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